Procedure and Methods: Sedov–Taylor + Gauss decreasing-density model and Migkas angular comparison

Working notes (Cosmology project)
Not a submitted methods section — living draft

18 September 2026

This document records the **ordered steps** of the model setup and calculations used in the present program: geometry, homologous expansion, Gauss gravity with decreasing density, spreadsheet / numerical evaluation, phase (axis) rotation, and comparison to Migkas et al. (2020) Fig. 6 longitude strips with uncertainties.

1 Scientific target and ontology

1. **Observational target.** Migkas et al. (2020) Fig. 6: best-fit L_X–T normalization $A/A_{\rm all}$ and slope $B/B_{\rm all}$ versus Galactic longitude ℓ in scanning strips of width $\Delta\ell = 90^\circ$ (full latitude), with uncertainties.

2. **Ontology (binding).** The Universe has a real geometric center of expansion (CoE) beyond the observable horizon. The observer sits off-center at Euclidean distance D from the CoE. The goal is to challenge the cosmological principle with a first-principles centered model, not to fit a free dipole template $A\cos\phi$ inside ΛCDM.

3. **Success criterion for the angular diagnostic.** After a simple sky-axis / phase alignment, the model longitude series should track *both* Migkas A and B bars (sheet diagnostic: $R^2 \sim 0.74$ for each). Separately, a pure ST+Gauss $H_{\rm obs}$ map must eventually pass a Migkas-like $\theta \simeq 75^\circ$ cone-scan filter (open).

2 Step 1 — Geometry

1. Place the CoE at the origin.

2. Place the observer at $\vec{D}$ with $D = \|\vec{D}\|$ and unit vector $\hat{D} = \vec{D}/D$ (CoE → observer).

3. Place a source at $\vec{r}$ (CoE-radius $r = \|\vec{r}\|$).

4. Define the observer–source separation

$$\vec{d} = \vec{r} - \vec{D}, \qquad d = \|\vec{d}\|, \qquad \hat{n} = \vec{d}/d.$$

5. On the spreadsheet / toy sky, sample sources on a circle of fixed observer-centric radius R about the observer:

$$\vec{r} = \vec{D} + R\,(\cos\varphi, \sin\varphi),$$

with Earth-centric angle φ measured from $+\hat{D}$ (so $\varphi = 0$ is anti-center, $\varphi = \pi$ toward the CoE). Tabulate $\varphi = 0°, 1°, \ldots, 359°$ (or equivalent).

6. Record Cartesian components P_x, P_y, then $r = \|\vec{r}\|$ and $d = R$ on that circle.

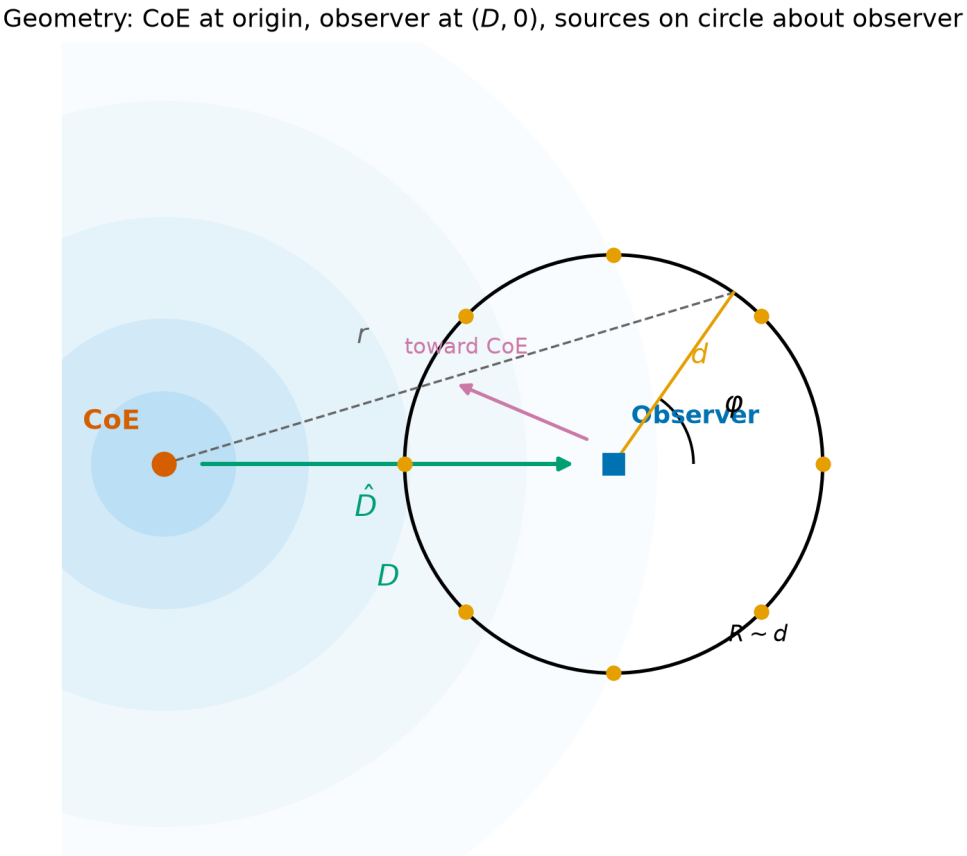


Figure 1: Geometry used in Steps 1–4: CoE, observer at D, source at r, separation d, Earth-centric angle φ.

3 Step 2 — Sedov–Taylor homologous background

1. Adopt a Sedov–Taylor (ST) blast-wave analogy with Hubble-like rate

$$H(t) \equiv \frac{\dot{R}_{\rm ST}}{R_{\rm ST}}.$$

(Cosmological map of blast energy and ambient density is left undefined; H is treated as an input scale.)

2. Interior homology (working assumption):

$$\vec{v}(\vec{r}, t) = H(t)\,\vec{r}.$$

3. Observer velocity $\vec{v}_O = H\vec{D}$; source velocity $\vec{v}_S = H\vec{r}$.

4. Relative recession (must subtract observer motion):

$$\vec{v}_{\rm rel} = H\vec{d}, \qquad v_{\rm los} = \vec{v}_{\rm rel}\cdot\hat{n} = H\,d, \qquad H_{\rm obs} = \frac{v_{\rm los}}{d} = H.$$

5. **Homology theorem (mandatory check).** For pure ST, $H_{\rm obs} = H$ is independent of direction for any D. If a numerical run shows angular variation at $M = 0$, the code failed to subtract $\vec{v}_O$ (frame error) and must be fixed before gravity is added.

4 Step 3 — Decreasing density and Gauss gravity

1. Reject uniform density: $M(< r) \propto r^3 \Rightarrow g \propto -r$ yields only a monopole in $H_{\rm obs}$ after differencing.

2. Adopt a centrally concentrated decreasing density whose cumulative mass is the incomplete-gamma form used in the spreadsheet (Phase A eq. 3.13):

$$\frac{m(r)}{M_{\rm tot}} = 1 - e^{-X}\Big(1 + X + \frac{X^2}{2}\Big), \qquad X = \frac{r}{L},$$

corresponding to $\rho(r) \propto e^{-r/L}$. Here L (sheet "density length") sets the concentration; binding runs use $D/L \sim \mathcal{O}(5\text{–}7)$.

3. Enclosed mass toward the CoE at the observer:

$$M_{\rm enc} \equiv m(D).$$

In the sheet, the labeled mass parameter is tied to $M_{\rm enc}$ (or to $M_{\rm tot}$ via the cumulative fraction); keep the identification explicit when changing units.

4. Gauss acceleration at CoE-radius r:

$$\vec{g}(r) = -\frac{G\,m(r)}{r^2}\,\hat{r}.$$

5. Peculiar-velocity ansatz (placeholder until full time integration from the blast):

$$\vec{u}(r) = -\alpha\,\frac{G\,m(r)}{r^2 H}\,\hat{r}, \qquad \alpha \sim \mathcal{O}(1).$$

(Intended physics: integrate $\vec{g}$ from the initial blast forward in time; the α/H form is the instantaneous O(1) surrogate.)

6. Relative LOS velocity with gravity:

$$\vec{v}_{\rm rel} = H\vec{d} + \big[\vec{u}(\vec{r}) - \vec{u}(\vec{D})\big], \qquad H_{\rm obs}(\hat{n}, d) = H + \frac{\delta v}{d}.$$

7. Natural dimensionless amplitude (binding):

$$\kappa \equiv \frac{GM_{\rm enc}}{D^3 H^2} = \frac{\Omega_m}{2} \approx 0.15 \quad (\Omega_m = 0.3).$$

This equates the mean enclosed density $\bar{\rho}(< D) = 3M_{\rm enc}/(4\pi D^3)$ with $\Omega_m \rho_{\rm crit}$ up to the factor that makes $\kappa = \Omega_m/2$. Local $\rho(D)$ in the exponential tail is *lower* than $\bar{\rho}(< D)$ (typically by $\sim$ 10–15$\times$ at $D/L \sim 6.7$); that is intentional for a decreasing profile, not a bug in κ.

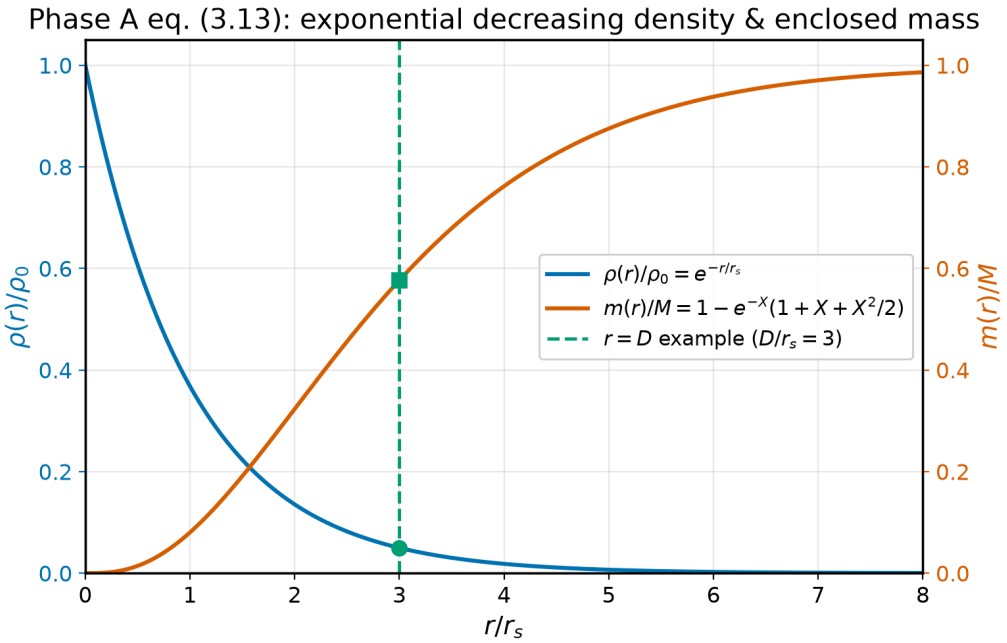


Figure 2: Illustrative decreasing $\rho(r)$ and cumulative $M(< r)$ used in Step 3.

5 Step 4 — Numerical evaluation on the sky circle

For each angle φ on the observer-centered circle of radius R:

1. Compute $\vec{r}$, r, $m(r)$, $\vec{g}$ or $\vec{u}$ at the source and at the observer.
2. Form ST velocities $\vec{v}_S = H\vec{r}$, $\vec{v}_O = H\vec{D}$ and peculiar difference $\vec{u}(\vec{r}) - \vec{u}(\vec{D})$.
3. Project onto $\hat{n}$ to obtain $v_{\rm los}$ and
$$\frac{H_{\rm obs}}{H} = \frac{v_{\rm los}}{H\,d}.$$
4. Store the series $H_{\rm obs}(\varphi)/H$ (sheet column historically tied to recession with acceleration over recession without).

5. Optional: map to an A-like quantity via luminosity-distance / flux scaling, e.g. $A \propto H^{-2}$,

$$\frac{A}{A_{\rm all}} \propto \frac{1}{(H_{\rm obs}/H)^2},$$

renormalised to mean unity over φ when comparing to Migkas $A/A_{\rm all}$.

Binding baseline parameters (unitless spreadsheet). Typical working set: $H \equiv k = 1$, $G = 1$, $D \sim 10$, $R \sim 2$–3, $L \sim 1.5$, $M_{\rm enc}$ chosen so $\kappa \sim 0.05$–0.15. Physical-unit copy tabs exist but are not trusted until G, H, and M are mutually consistent with $\kappa \sim \Omega_m/2$.

6 Step 5 — Migkas data: digitise Fig. 6 bars

1. Obtain Migkas et al. (2020) Fig. 6 (top $A/A_{\rm all}$, bottom $B/B_{\rm all}$ vs Galactic longitude).
2. Digitise central values and asymmetric error bars on the $\Delta\ell = 90^\circ$ strip series.
3. Store as CSV with columns

$$\ell, \quad A/A_{\rm all}, \quad A^{\rm lo}_{\rm err}, \quad A^{\rm hi}_{\rm err}, \quad B/B_{\rm all}, \quad B^{\rm lo}_{\rm err}, \quad B^{\rm hi}_{\rm err}.$$

(Present file: `migkas/fig6_digitized.csv`, $\sim 5^\circ$ sampling, 71 points.)

4. Always plot Migkas points **with error bars** in comparison figures (same presentation as the published figure).

7 Step 6 — Phase (axis) rotation diagnostic

The spreadsheet "rotation" is a **simple sine phase shift** of the angular template (not a 3D geometric re-solve of the CoE for each trial, and not a free dipole MCMC).

1. Define

$$S(\ell;\phi) = A_{\rm mp} \sin\Big(\frac{2\pi}{360}\,\ell + \phi\Big),$$

with amplitude $A_{\rm mp}$ and phase ϕ in radians (sheet: $\omega = 2\pi/360$ so ℓ is in degrees).

2. Binding sheet values (tab *Copy of Copy of Copy of density*):

$$A_{\rm mp} = -0.06, \qquad \phi = 0.35 \text{ rad} \approx 20.05^\circ.$$

Opposite amplitude for B: $B_{\rm mp} = -A_{\rm mp}$.

3. Construct rotated model series from the ST columns (sheet notation):

$$\text{Model}_A = \big(1 - S_A(\ell;\phi)\big) \times (\text{ST } A\text{-like series}),$$
$$\text{Model}_B = \big(1 - S_B(\ell;\phi)\big) \times (\text{ST } B\text{-like series}).$$

4. Align model longitudes to Migkas ℓ (sheet: `VLOOKUP` / interpolation onto the Migkas angle grid).

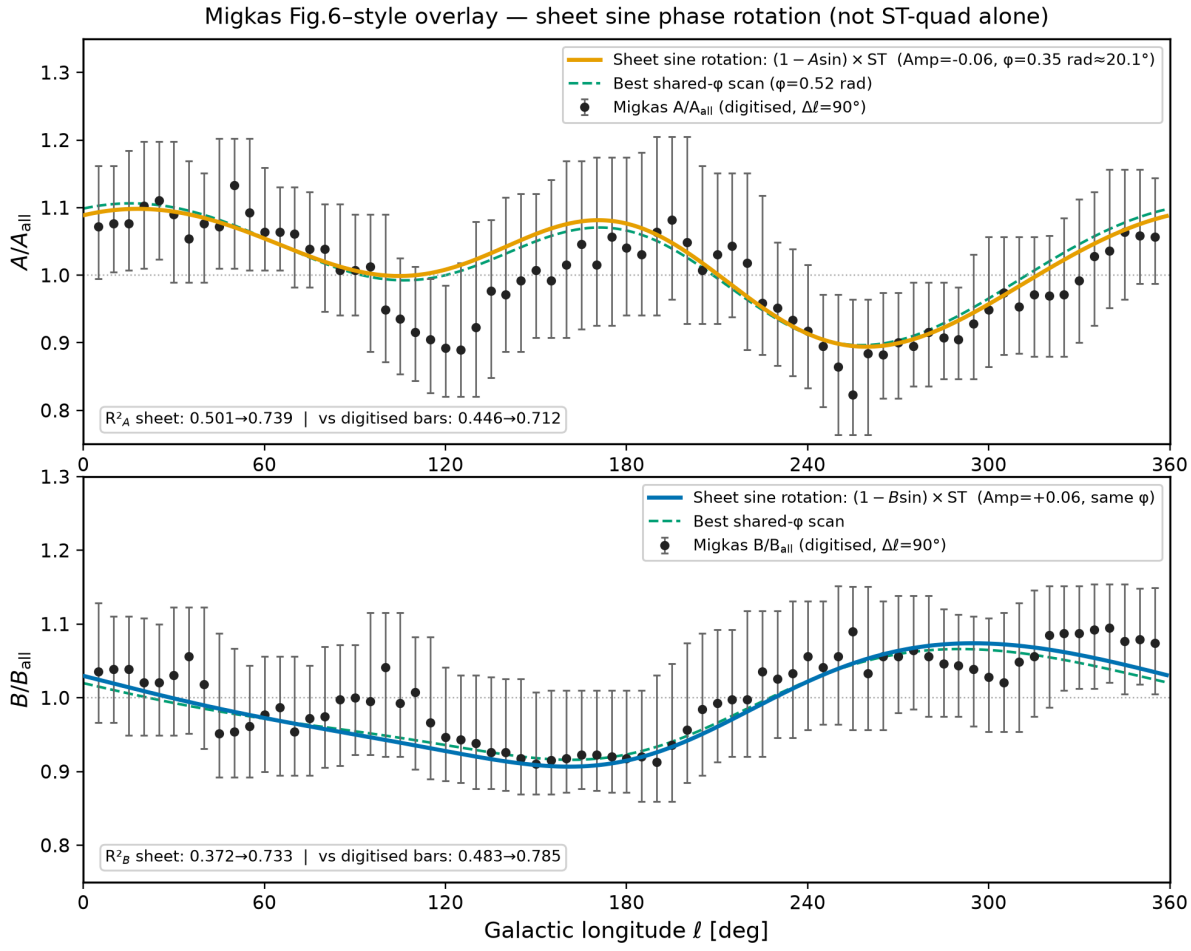
5. Compute ordinary coefficients of determination R^2 for Migkas vs model, *with* and *without* the sine phase factor. Report the ratio $R^2_{\text{with}}/R^2_{\text{without}}$ (sheet cells historically `R2w/R2wo`).

6. Optional: scan shared ϕ to maximise a joint $A{+}B$ score; record best ϕ separately from the author's fixed $\phi = 0.35$.

Recorded results (18 Sep 2026).

Series	R^2 without rotation	R^2 with sheet rotation
A/A_{all} (sheet 1° grid)	0.501	**0.739**
B/B_{all} (sheet 1° grid)	0.372	**0.733**
A vs digitised CSV+bars	0.446	0.712
B vs digitised CSV+bars	0.483	0.785

Best shared-ϕ scan vs digitised bars: $\phi \approx 0.52$ rad ($R^2_A \approx 0.730$, $R^2_B \approx 0.776$).



Diagnostic: Amp×sin((2π/360)·ℓ+φ) phase alignment from Chris sheet tab "Copy of Copy of Copy of density". ST quadrupole alone is a separate claim.

Figure 3: Step 6 output: Migkas Fig. 6-style bars for A and B with sheet sine-rotation models overlaid.

8 Step 7 — Fit diagnostics (required plots)

1. **Longitude overlay:** Migkas A and B vs ℓ with error bars; model curves after phase rotation on the same axes.
2. **Scatter:** Migkas vs model for A and for B, with a plain linear fit $y = ax + b$ and large R^2.
3. **Rotation improves fit:** R^2 versus phase ϕ (or sky-axis angle ϕ_0), marking the author's ϕ and any prior related to Migkas's preferred longitude.

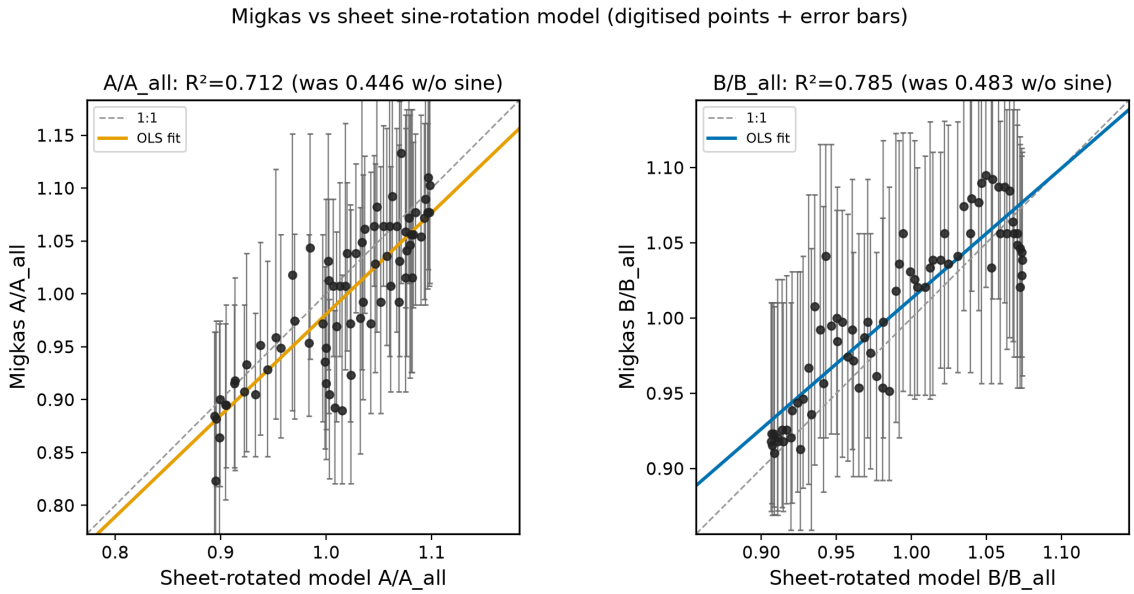


Figure 4: Step 7 scatter diagnostics for A and B after phase rotation.

9 Step 8 — Sanity and go / no-go checks

1. $M = 0$ (or $\alpha = 0$): $H_{\rm obs}(\varphi)$ must be flat.
2. Uniform ρ: sky anisotropy must collapse relative to decreasing ρ.
3. κ from sheet $(G, M_{\rm enc}, D, H)$ must be $\mathcal{O}(\Omega_m/2)$; reject unit systems that yield absurd κ.
4. Both A and B must be shown with Migkas bars; do not report B as a "null" under the sheet rotation diagnostic.
5. Separate claim (weaker until cone-scan exists): raw ST+Gauss quadrupole shape alone vs B without Amp$\times$ sin phase factor.

10 Step 9 — Open methods (next, not yet locked)

1. Forward-model $H_{\rm obs}(\hat{n})$ through Migkas-like $\theta \simeq 75^\circ$ sky cones / longitude strips (same pipeline as the paper), then retune α, κ only against that filter.

2. Replace the α/H peculiar ansatz by time-integrated gravity from the blast.

3. Light-travel / luminosity-distance composition (Newtonian $v_{\rm los}/d$ is the present pass only).

4. CMB multipole adversary for an off-center observer with a beyond-horizon CoE.

11 Summary flowchart (calculation order)

Step	Action
1	Fix CoE, $\vec{D}$, sky circle (φ, R)
2	Homologous ST; verify isotropic $H_{\rm obs}$ at $M = 0$
3	Decreasing ρ; Gauss g; $\kappa = GM_{\rm enc}/(D^3H^2)$
4	Compute $H_{\rm obs}(\varphi)$; optional $A \propto H^{-2}$ map
5	Digitise Migkas Fig. 6 A, B with error bars
6	Apply Amp$\times \sin(\omega\ell + \phi)$ phase rotation; compute R^2
7	Overlay, scatter, R^2 vs phase plots
8	Sanity / unit / both-A-and-B checks
9	Cone-scan, time-integrated g, light travel, CMB (open)

References

[1] K. Migkas et al., Astron. Astrophys. **636**, A15 (2020), arXiv:2004.03305.